



Travis Association for the Blind

3PL Enablement Design Document

Austin, TX

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INTRODUCTION

This document is designed to outline the proposed 3PL enablement process for Supply Chain Architected for Logistics Execution (SCALE) at Travis Association for the Blind. The functional design outlined in this document is intended solely for use at the specified Travis Association for the Blind facility in Austin, TX. Future implementations at any other Travis Association for the Blind facilities will result in either change to this document or entirely new functional flows per facility.

The objective of this document is to define the systemic and functional processes and scope from a Manhattan Associates' (MA) SCALE perspective. This document serves as a reference throughout the 3PL enablement process for confirmation of approach and definition of tasks. It will also serve as a reference for Manhattan Associates' customer support organization (CSO) after implementation, and potentially a reference point for any future Travis Association for the Blind implementations.

IMPORTANT NOTE

Travis currently operates as a 3PL for a single customer (DLA). This customer has some contractual restrictions regarding physical access to their merchandizing. Based on the current contract, Travis would need to have a physical segregation (fence/wall) to keep DLA's operation apart from any other operation at Travis' warehouse.

Travis is not envisioning to enable another 3PL operation in the near future, mainly because of the requirements of their current customer.

This document will outline the requirements and recommendations to enable another 3PL.

STATISTICS

Basic statistics of the Austin, TX Warehouse can be found below:

- **General Information**
 - ~480,000 sqft.
 - 1 shift (5am ~ 3:30pm / Mon~Fri)
 - Travis may have overtime or additional operation days if needed
- **Outbound Volume**
 - ~30k shipments per month
 - The United States Marines and United States Air Force are the main customers for Travis
- **Inbound**
 - ~4,000 pallets per month

WAREHOUSE LAYOUT

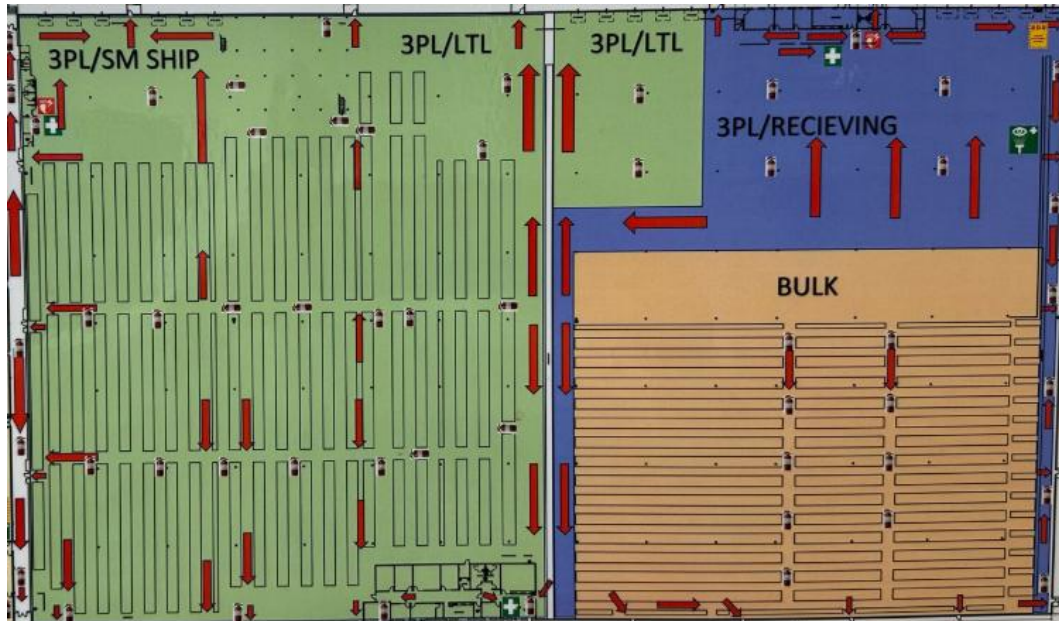


Figure 1 - Warehouse layout

TECHNOLOGY

Host: Rainbow Data Systems

Middleware: Dell Boomi

WM Platform: Windows

Version: Active SCALE

RF Vendor: Zebra

MHE Vendor: **Pack Size and EMIT Inc**

Label Printer: Zebra (203 dpi)

Chrome Version: **TBD**

KEY DECISIONS / ASSUMPTIONS

1. Travis will create a Company for each new 3PL customer added to their operation.
2. The Item Master will have the Company.
3. The SKU will be unique across companies. This is not a SCALE limitation, but having multiple companies with the same SKU may be a challenge for the users during work execution due to the use of VoCollect.
4. The Storage Template will be defined depending on the item's characteristics.
5. The dimensions/weight will be maintained at the Unit of Measure level.
6. The Treat as Loose flag will be maintained at the Unit of Measure level.
7. Ideally Travis would need to configure new zones (locating, allocation and work) to keep the customers' inventory apart from other customers.
8. Travis may use permanent locations for items with high demand at picking locations.
9. A new set of locating configuration (Locating rule assignment criteria, Locating rule assignment, Location selection and Locating rule) is required to be created per customer, based on the zones and companies.
10. A new set of allocation configuration (Allocation rule assignment criteria, Allocation rule assignment, Allocation location selection and Allocation rule) is required to be created per customer, based on the zones and companies.
11. Travis may create a new Wave flow and Wave master with specific requirements for the customer. Travis can also opt to use a standard Wave flow/master for all customers, as long as the wave flow steps are not company specific.
12. The Item Master, Receipt Download and Shipment Download should be downloaded into SCALE by Travis' middleware provider. This is a Manhattan Associates recommendation to keep the logic to convert the data provided by the customer into a format that SCALE recognizes (XML).
13. The Item Balance Upload, Inventory Transactions Upload, Receiving Upload and Shipping Upload should be generated as XML files by SCALE into separate folders, one per customer, at the Azure Storage. From there, Travis' middleware should pick the files and convert the information into the format that Travis' customer requires.
14. In case the customer requires the use of Lot, Travis will be able to active lot tracking at the Item Master.
 - a. Travis understands that the use of lot tracked items represents a change to their current operation that could affect Receiving, Picking, Replenishment, Cycle Count, Inventory Management, Packing, Outbound QC, etc.
 - b. SCALE is fully prepared to work with lot tracked items, but the operational changes need to be evaluated by Travis to define new SOPs.
15. Travis will test the Blind Receiving available in Warehouse Mobile on the current SCALE's version (24.7.2575). When the original design was made this feature was not yet available.
16. Travis will test the Shipping Container QC available in Warehouse Mobile on the current SCALE's version (24.7.2575). When the original design was made this feature was not yet available.
17. Travis will test the Status Change available in Warehouse Mobile on the current SCALE's version (24.7.2575). When the original design was made this feature was not yet available.
18. Other configurations and processes not mentioned in this document should be kept as originally designed in the Solution Design Document.

TERMINOLOGY

Client Terminology	WM Terminology	Definition
	NMFC	NATIONAL MOTOR FREIGHT CLASSIFICATION (LTL CLASS). NMFC IS UTILIZED ON BILL OF LADING DOCUMENT TO DISPLAY THE NMFC CLASS OF THE FREIGHT.
	WAVE	A WAVE REPRESENTS THE DIFFERENT STEPS THAT THE SYSTEM USES TO RETRIEVE ORDERS FROM THE POOL, AND PROCESS THEM INTO THE OUTBOUND PORTION OF SCALE
	OVERRIDE DATA WAVE STEP (ODWS)	THIS WAVE STEP ALLOWS YOU TO UPDATE DATA (THAT YOU NORMALLY WOULD NOT HAVE ACCESS TO) DURING THE WAVE PROCESS
	STATUS FLOW	A STATUS FLOW IS A DEFINED SEQUENCE OF STATUSES GROUPED TOGETHER TO MANAGE PROCESSING
	LPN	LICENSE PLATE NUMBER
	VAS	ITEMS THAT REQUIRE SPECIAL HANDLING
	P&D	PICK UP AND DROP LOCATION
	UI	USER INTERFACE
	ASN	ADVANCE SHIPPING NOTIFICATION
	PO	PURCHASE ORDER
	HAZMAT	HAZARDOUS MATERIAL
	EXIT POINT	EXTERNAL PROCESS THAT PERFORMS LOGIC IN LINE OF THE BASE PROCESS. USED TO UPDATE DATA OR PERFORM ADDITIONAL LOGIC.
	MIF	MANHATTAN INTEGRATION FRAMEWORK
	UDF	USER DEFINED FIELD
NSN	ITEM	

I. INTERFACES

Travis downloads & uploads information into SCALE using XML file-based methods of interfacing.

This allows Travis to create records with key information for downloads, and SCALE reads this key information to process the download into the SCALE production tables. This method also allows Travis to read key information from SCALE's upload interface records.

Each interface touch point below can be run manually as well as through scheduled jobs as defined by Travis Association for the Blind. The specific schedule can depend on the Host System, warehouse processing times, and SCALE interface execution times.

Warehouse alerts can be configured to notify Travis when a download or upload interface transaction has failed for any reason.

1.0 DOWNLOAD FROM HOST TO SCALE

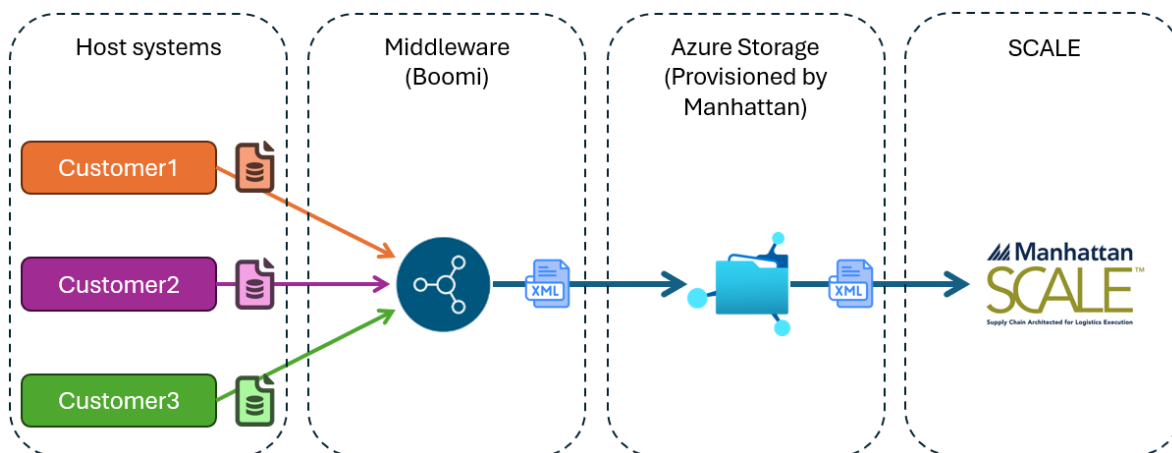


Figure 2 - Download touchpoints

1.1 Download data

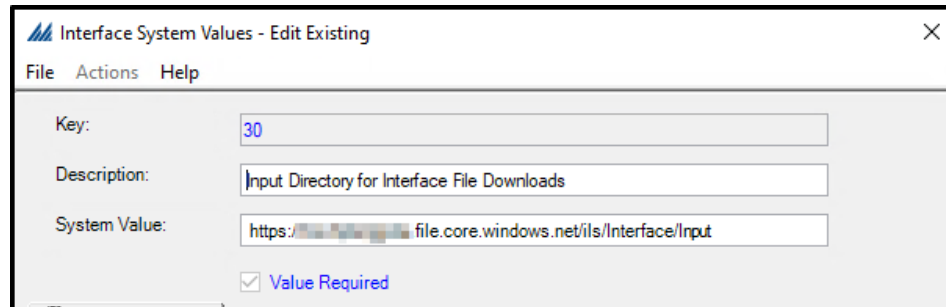
Travis will download the following data into SCALE:

- o Item Master
- o Receipt Download
- o Shipment Download

1.2 Download architecture

Travis will use their middleware (Boomi) to download information to SCALE. Boomi must receive/pull the necessary information from Travis' customers, translate that information into SCALE's format and then generate the expected XML file.

The downloaded files will be stored in a single Azure Storage folder (/ils/Interface/Input).



Interface System Values - Edit Existing

File Actions Help

Key: 30

Description: Input Directory for Interface File Downloads

System Value: https://file.core.windows.net/ils/Interface/Input

☒ Value Required

Figure 3 - Interface download folder

1.3 Item Master

The item master file is maintained within the Host. Conversion quantities, weights, and dimensions are also maintained through this Item Master Download as those pieces of information are available in the host. The middleware receives information from the host system and downloads a record into the SCALE Item Master when an item is modified or created. If an item already exists in the Item Master, then the record is flagged as a change, and SCALE updates the existing item with the latest information bridged from the host. Key pieces of information include (but are not limited to) the following:

- o Item Master Level:
 - Item (SKU/NSN)
 - Description
 - Company
 - Cross Reference
 - Packing Class
 - Item Class
 - Storage Template
 - Item Unit of Measure Level:
 - Item Dimensions
 - Item Weight
 - Movement Class (A, B, C)
 - Item Categories

Note: SKUs at Travis are referred to internally as NSNs and this terminology may be used throughout the remainder of this document.

Config Note: To be able to use the XML download interface option, the 'Item XML Download' must be enabled under the 'Interface Process' configuration.

1.4 Receipts

Receipts should be downloaded into SCALE in advance, so the users can receive against an existing receipt. Travis may also use Blind receiving for unexpected returns, although this process requires previous agreements with Travis' customers.

Note: The receipts must have the Company field interfaced at the header level.

The information downloaded to SCALE from Travis will always contain both header and detail level information. Container Level Receipt data, if available from customer, could also be interfaced in advance to speed up the check-in process.

SCALE Table	Description
Receipt Order Header	Header Level Receipt Data
Receipt Order Detail	Line-Item Detail Receipt Data
Receipt Order Container	Container level receipt data (if available from customer)

1.5 Shipments

The shipments information will flow through Boomi which will be responsible for any conversion/formatting required. Once formatted, the Shipments will be passed onto SCALE via XML file-based interfaces. Shipment data is sent with a warehouse-specific shipment header, one or more shipment details, and optional shipment comments. The shipment header contains information such as customer, customer address, ship to address, carrier, and scheduled ship date information. The shipment detail is the item level detail for the shipment containing information such as item and ordered quantity. Comments can be linked to the shipment header or details to validate additional processing requirements within the warehouse. Until the shipment has been waved, SCALE can process updates and deletions from the host against the shipment.

Note: The shipments must have the Company field interfaced at the header level.

SCALE Table	Description
Shipment Header	Header Level Shipment Data
Shipment Detail	Line-Item Detail Data
Shipping Container	NOT DOWNLOADED – Container Level Data
Shipment Comment	Header or Detail Level Comments

2.0 UPLOAD FROM WM TO HOST

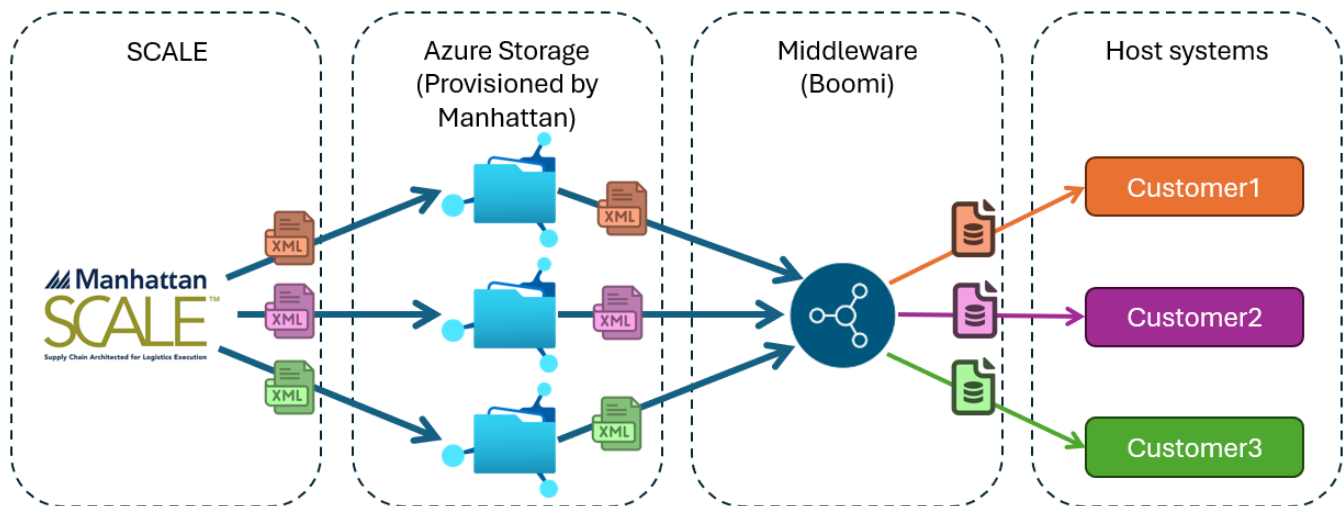


Figure 4 - Upload touchpoints

2.1 Upload data

SCALE will upload the following data:

- o Receipt Confirmation
- o Shipment Confirmation
- o Inventory Transactions
- o Item Balance

2.2 Upload architecture

SCALE will generate the upload data into different Azure Storage folders, based on the Interface Process configuration. These configurations should leverage base SCALE criterias to generate the files into the corresponding customer's folder.

From the Azure Storage, middleware (Boomi) should read the information per folder and convert the information back to customer's host systems.

2.3 Receipt Confirmation

The Interface Data option creates the receipt upload files from SCALE for all receipts that have reached the trailing status 'In Putaway' or have been closed manually. The receipt upload files are the output of the receiving processes within SCALE. The upload includes receipt header, receipt detail, and receipt container level information. As each receipt container reaches 'In Putaway' status (Container Level Upload), the interface sends information to the host system to allow that receipt's inventory to be acknowledged as present in the facility.

For Travis Association for the Blind, the upload will be performed at **Container** level when the status of the receipt reaches 'In Putaway'.

Config Notes:

A new **Receiving interface upload criteria** should be created and configured for each customer, using the Company as the filter attribute.

Receiving interface upload criteria - Edit Existing

File Actions Help

Record Type: RECINTUPLCRIT

Filter Name: CUST1

Description: CUST1

Table name: Receipt Header

Filter data

Filter Criteria

☒ And ☐ Or

Attribute: Operand: Value:

Receipt_Header.COMPANY = 'NCUST1'

Add Rule

A new **Interface Process Detail** should be created and configured for each customer, using the criteria created in the previous step. It should also indicate an exclusive folder per customer.

Interface Process Detail - Create New

File Actions View Help

Sequence: 801

Description: Receiving XML Upload - CUST1

General Upload Flow Control User Defined Data

Upload directory: https://file.core.windows.net/ILS/Interface/Upload/CUST1

Upload criteria

Shipping: Shipping delete criteria:

Receiving: CUST1 Item Balance:

Inventory Transaction:

2.4 Shipment Confirmation

The Interface Data option creates the shipment upload files from SCALE for all shipments that have reached status 'Closed' (shipment has been loaded to the truck and truck has left the warehouse). The shipment upload files are the output of the outbound process within SCALE. These files are generated after the execution of the Shipping Load confirmation. The upload includes shipment header, shipment detail, shipment comment, and shipping container information.

For Travis Association for the Blind, Shipment uploads are created only at **Closed** (900) status.

Config Notes:

A new **Shipping interface upload criteria** should be created and configured for each customer, using the Company as the filter attribute.

A new **Interface Process Detail** should be created and configured for each customer, using the criteria created in the previous step. It should also indicate an **exclusive folder per customer**.

2.5 Inventory Transactions

SCALE maintains 4-wall inventory at a detailed level and communicates any changes in inventory levels to Travis through the Inventory Transactions Interface. All movements and changes in inventory are eligible to be uploaded (Adhoc adjustments, cycle count adjustments, status changes, etc).

Travis can use the base configuration in SCALE to not interface specific transactions, based on their business requirements. To achieve that, the Adjustment Type used to generate the inventory transaction should have the field Include in interface uploads unchecked.

Adjustment Type - Edit Existing

File Actions Help

Adjustment Type: Transfer-No Work Creation

Description: Inventory Transfer - No Work Creation

General Authorized Users Warehouse Access User Defined Data

Adjustment Class

☐ Adjustment
☐ Status Change
☐ Company Transfer
☒ Transfer
☐ Warehouse Transfer

Quantity Values

Minimum Qty: 0.00000

Maximum Qty: 999999.00000

Initiation type

☒ Item/Location
☐ License plate

Work Values

☐ Create Work
 Work Creation Master:

☐ Allow for frozen locations.
☒ Include in interface uploads.

Config Notes:

A new **Inventory transaction interface upload criteria** should be created and configured for each customer, using the Company as the filter attribute.

Inventory transaction interface upload criteria - Edit Existing

File Actions Help

Record Type: INVENTORYINTUPLOADCRT

Filter Name: CUST1

Description: CUST1

Table name: Transaction History

Filter data

Filter Criteria

☒ And
☐ Or

Attribute: Transaction_History.COMPANY

Operand: =

Value: N'CUST1'

Add Rule

A new **Interface Process Detail** should be created and configured for each customer, using the criteria created in the previous step. It should also indicate an exclusive folder per customer.

Interface Process Detail - Create New

File Actions View Help

Sequence: 701

Description: Inventory Transaction XML Upload - CUST1

General Upload Flow Control User Defined Data

Upload directory: https://file.core.windows.net/ils/Interface/Upload/CUST1

Upload criteria

Shipping: Shipping delete criteria:

Receiving: Item Balance:

Inventory Transaction: CUST1

2.6 Item Balance

The total on-hand inventory for a given item and inventory status can be uploaded to the host system on-demand through the Interface Data option in SCALE, or on a scheduled basis as configured in the Scheduled Job option. Any time the total on hand quantities are needed to compare against the host system, this option can be run and sent to the host for reporting & comparison purposes.

Config Notes:

A new **Item balance interface upload criteria** should be created and configured for each customer, using the Company as the filter attribute.

Item balance interface upload criteria - Edit Existing

File Actions Help

Record Type:

Filter Name:

Description:

Table name:

Filter data

Filter Criteria

Attribute	Operand	Value
LOCATION_INVENTORY.COMPANY	=	'NCUST1'

Add Rule

A new **Interface Process Detail** should be created and configured for each customer, using the criteria created in the previous step. It should also indicate an exclusive folder per customer.

Interface Process Detail - Create New

File Actions View Help

Sequence:

Description:

General Upload Flow Control User Defined Data

Upload directory:

Upload criteria

Shipping:		Shipping delete criteria:	
Receiving:		Item Balance:	CUST1
Inventory Transaction:			

II. INBOUND

3.0 LOCATING RULES

In order to select the putaway locations based on the Company, a new set of configurations must be created in SCALE. These configurations will assign a locating rule based on a criteria. Multiple configurations may need to be created for each customer to accommodate the requirements of storage of products and the unit of measure being received.

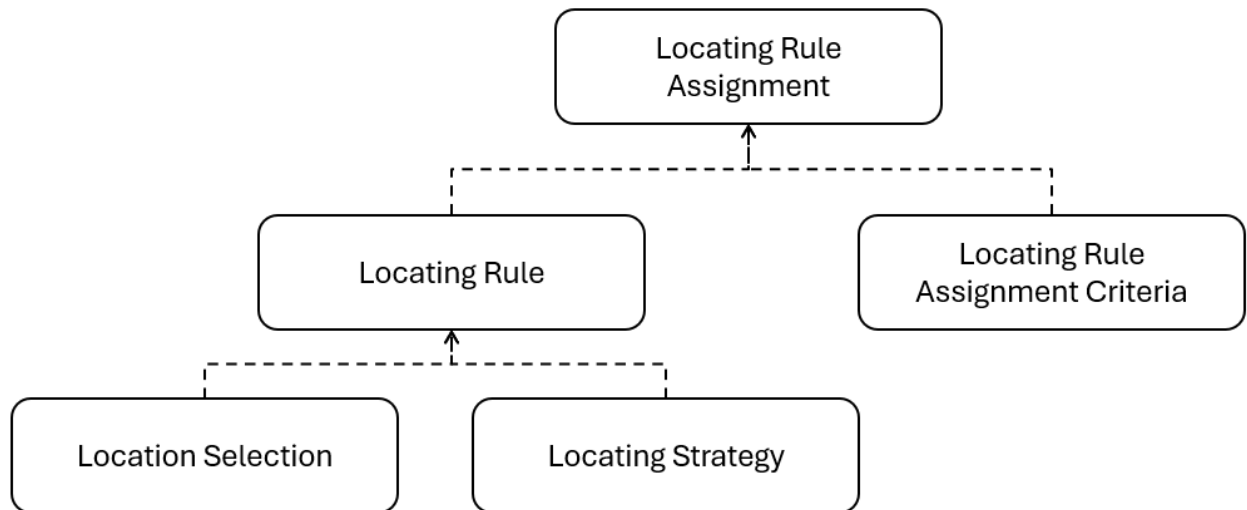


Figure 5 - Locating configuration

3.1 Location selection

One or more Location selection configuration must be created. The location selection will indicate which locations should be evaluated during the locating process. Usually there are a few location selections based on the zones and the unit of measure that should be stored in that zone (EA, CS, PL).

3.2 Locating rule

The locating rules must be created according to the requirements to store the items received. A locating rule may have several detail records with different location selections and strategies.

Sequence	Locating Strategy	Location Selection	Split Quantity?
10	Consolidate to location that already contains t...	CUST1	No
20	Empty location	CUST1	No
30	Use specific (multi-item) location regardless of...	SUPERVISOR	No

3.3 Locating rule assignment criteria

The criteria must use the company at the receipt detail as a filter.

RECEIPT_DETAILCOMPANY = NCUST1

Add Rule

3.4 Locating rule assignment

This is the configuration that will tighten together the locating rule and the criteria, to indicate to SCALE which locating rule should be used during the check-in.

Priority: 310

Criteria: CUST1

Locating Rule: CUST1

Always Override

III. INVENTORY CONTROL

4.0 REPLENISHMENT

Replenishment uses allocation rules to identify the inventory that should be allocated. These rules should use the zones defined for each customer.

5.0 CYCLE COUNT

Both the Cycle Count Master and Cycle Count Quick Plan can leverage the use of the zones and/or company to filter the inventory that should be counted.

The frequency of the cycle count depends on the contract with the 3PL customer, so its recommended to create the configurations for each customer to better accommodate any requirements changes.

IV. OUTBOUND

6.0 ALLOCATION RULES

In order to select the inventory that is going to be shipped based on the Company, a new set of configurations must be created in SCALE. These configurations will assign an allocation rule based on a criteria. Multiple configurations may be created to accommodate the business requirements during the outbound process.

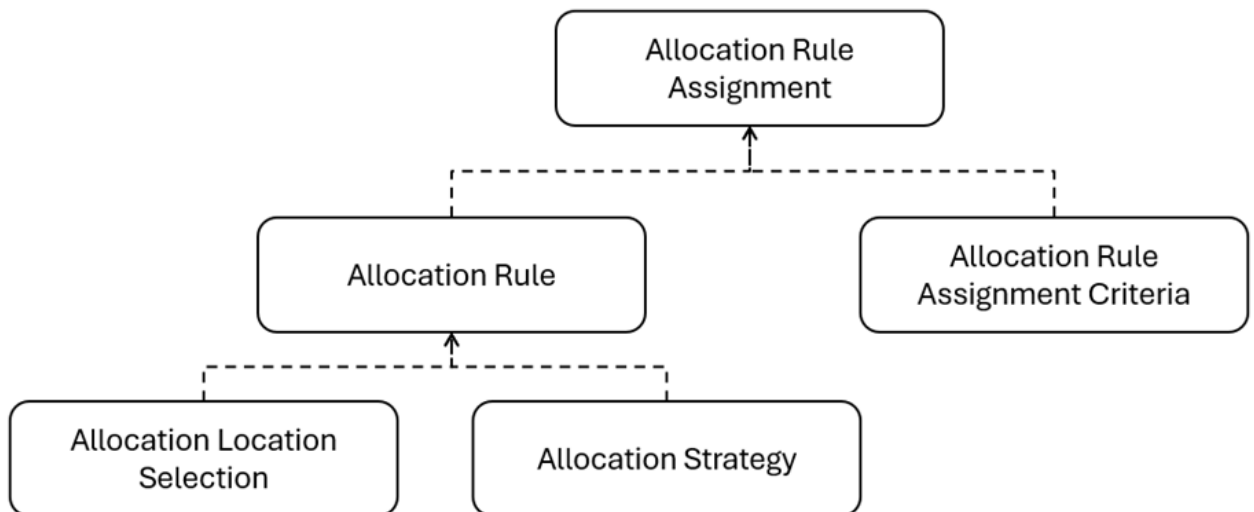


Figure 6 - Locating configuration

6.1 Allocation location selection

One or more Allocation location selection configuration may be created. The allocation location selection will indicate which locations should be evaluated during the allocation process. Usually there are a few allocation location selections based on the zones and the unit of measure that should be stored in that zone (EA, CS, PL).

6.2 Allocation rule

The allocation rules must be created according to the requirements to assign the inventory to the orders. An allocation rule may have several detail records with different allocation location selections and strategies.

Sequence	Strategy	Location Selection	Eligible ums	Lot
10	CUST1	CS,EA,PL		

6.3 Allocation rule assignment criteria

The criteria must use the company at the shipment detail as a filter.

Filter Criteria

And Attribute: Operand: Value:

COMPANY = N'CUST1'

Add Rule

6.4 Allocation rule assignment

This is the configuration that will tighten together the allocation rule and the criteria, to indicate to SCALE which allocation rule should be used during the waving process.

Priority: 93

Criteria: CUST1

Allocation Rule: CUST1

Always Override

6.5 Rule Assignment

To make the above configuration work, it is important to add the wave step Rule Assignment to the wave flows. This wave step will evaluate the configurations defined above and will assign the corresponding Allocation Rule to the shipment lines.

This wave step needs to be placed **before** the Allocation wave step.

Wave Flow - Edit Existing

File View Actions Help

Wave Flow: STZ-LTL

Description: STZ-LTL

Detail Records | User Defined Data

Sequence	Description
10	Start Wave
20	Reset Alloc Rule, Criteria and Status Flow
30	Set LTL Carrier
40	Rule Assignment
50	Allocation
60	Container Creation
65	OC Assignment

7.0 WAVE MANAGEMENT

Travis can keep using the same Wave Flows and Wave Masters as they use today. In case special requirements from a new 3PL customer is needed, Travis can request to make changes to existing Override Data Wave Step or to create new ones to accommodate the business requirements.

8.0 WORK MANAGEMENT

Travis may need to define different priorities based on customers' requests. Travis already deals with orders/work that requires an elevation on the priority.

Travis will need to develop a SOP for this process.

9.0 PICKING

Travis may continue using the current Work Types for most of their picking processes. In case they need to have specific people working on different zones, this can be achieved by using the Work Profile's Work Zone configuration.

10.0 PACKING

10.1 VAS (Value Added Services)

The criteria used to assign VAS to the containers may also leverage the Company as a filter. In case Travis needs to add VAS to specific customers only, a new VAS Activity Criteria needs to be created and used on the VAS Activity screen.

The VAS Assignment wave step needs to be added after the Container Creation wave step, so SCALE can evaluate the containers based on the VAS Activity Criteria. Multiple criterias can be created. SCALE will evaluate all the active criterias during the VAS Assignment wave step.

Sequence	Description
41	Hard update allocation rule to STZ
50	Allocation
60	Container Creation
63	VAS Assignment
65	QC Assignment
70	Set LTL status flow in shipment details

Note: VAS can be assigned only when containers are created in Wave.

10.2 QC Process (Order Verification/OV)

Similar to VAS, the QC Assignment Criteria and the QC Assignment can be created to accommodate requests from 3PL customers related to the Outbound QC verification.

Note: QC can be assigned automatically only when containers are created in Wave.

REVISION HISTORY

Date:	Changed By:	Doc. Version:	Notes:
03/05/2025	Luciano Peres	1.0	Original Document

ACKNOWLEDGEMENT of 3PL Enablement Design Document – Version 1.0

Signature: _____ **Date:** _____